



UNIVERSITY OF HELSINKI
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Dynamic visual search modelled with foveated observer

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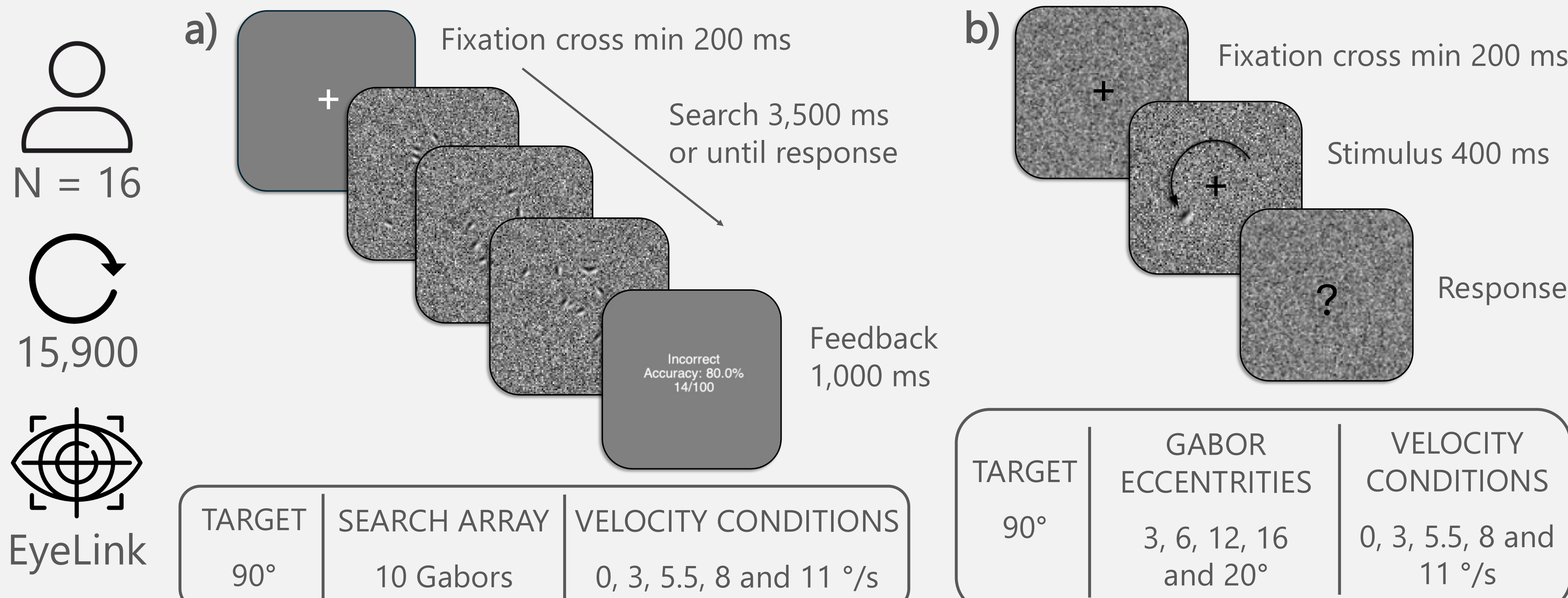
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Background

Our everyday visual environment is dynamic and constantly changing. While a great deal is known about how we gather visual information in static settings (Eckstein et al., 2006; Wolfe & Horowitz, 2017), dynamic environments remain less studied.

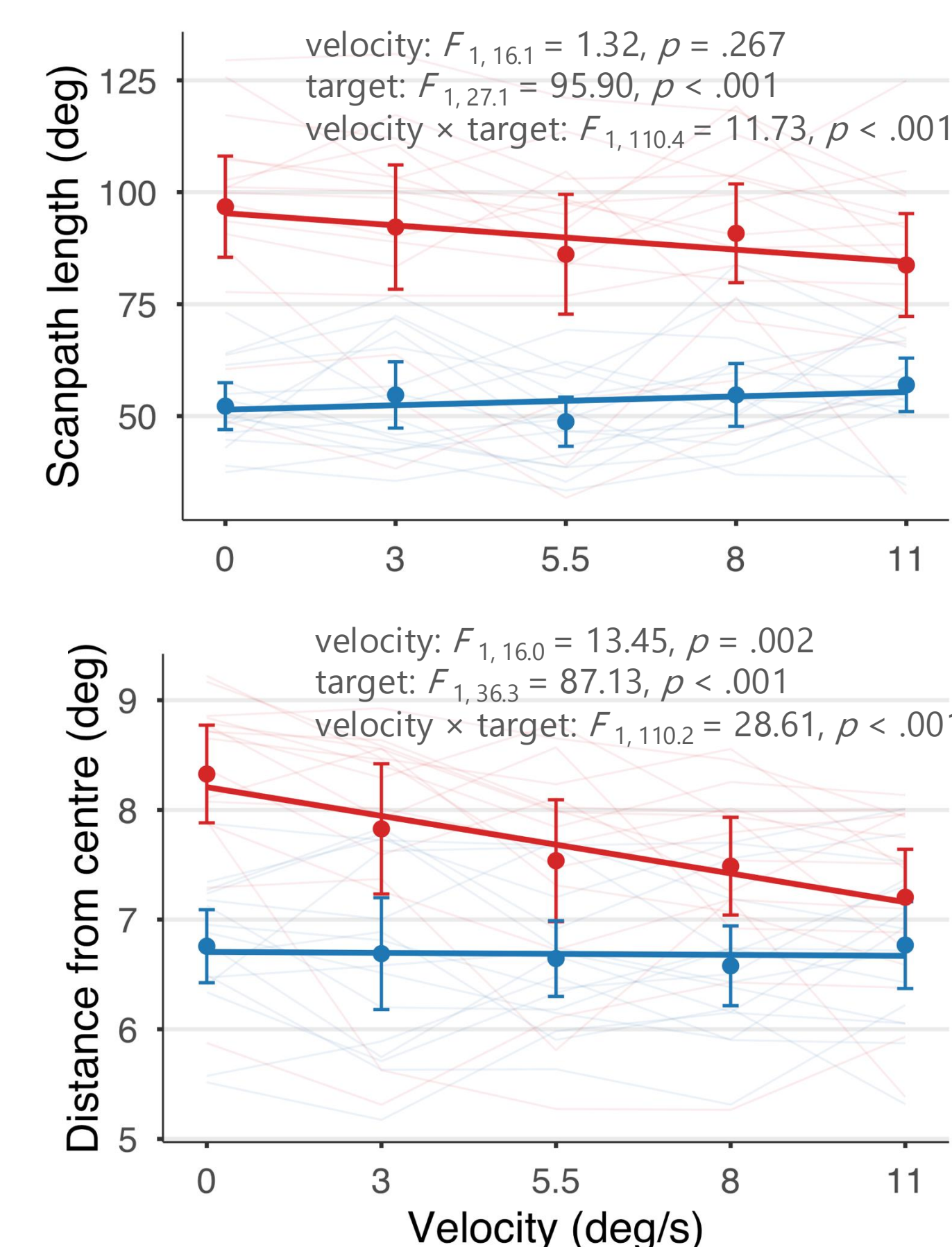
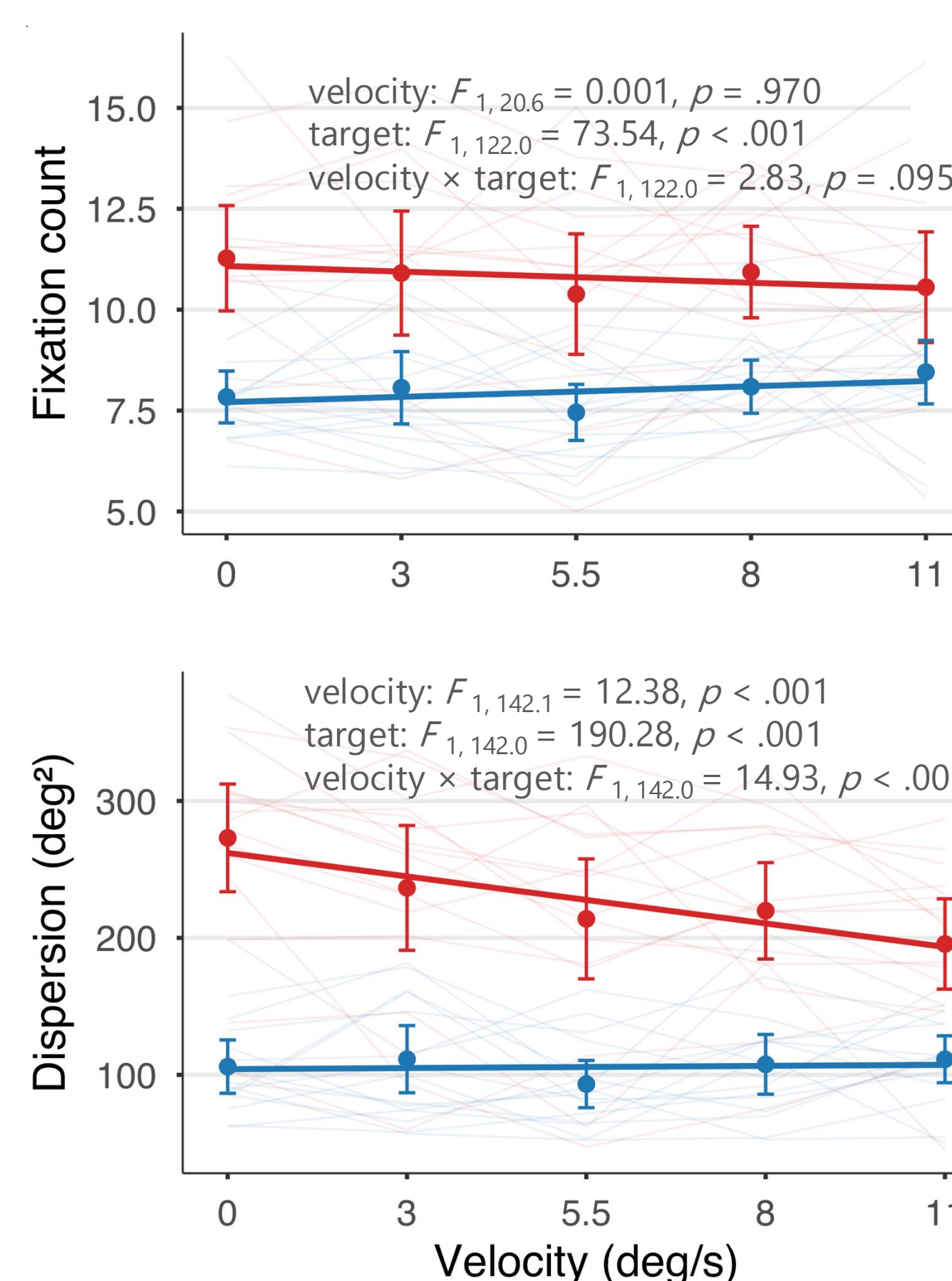
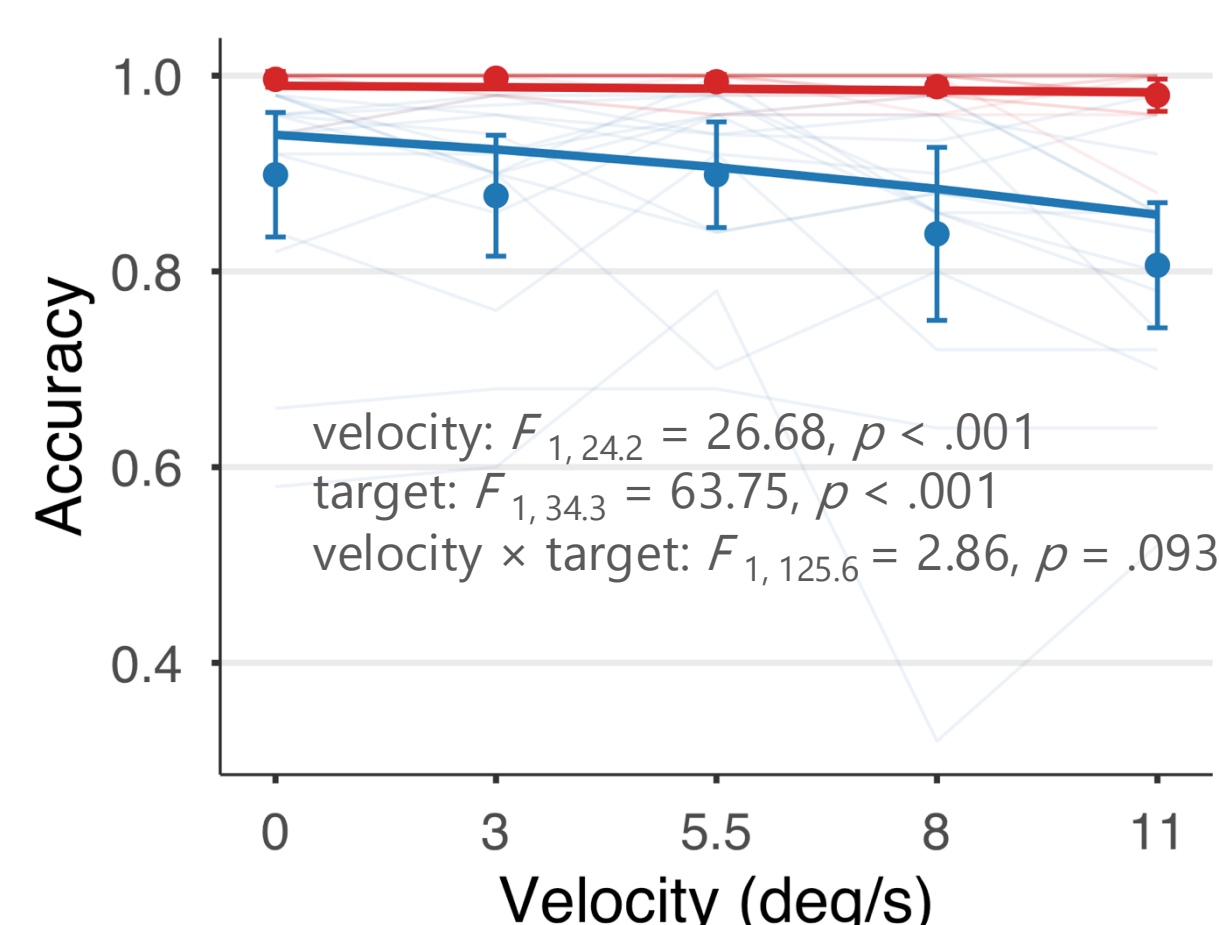
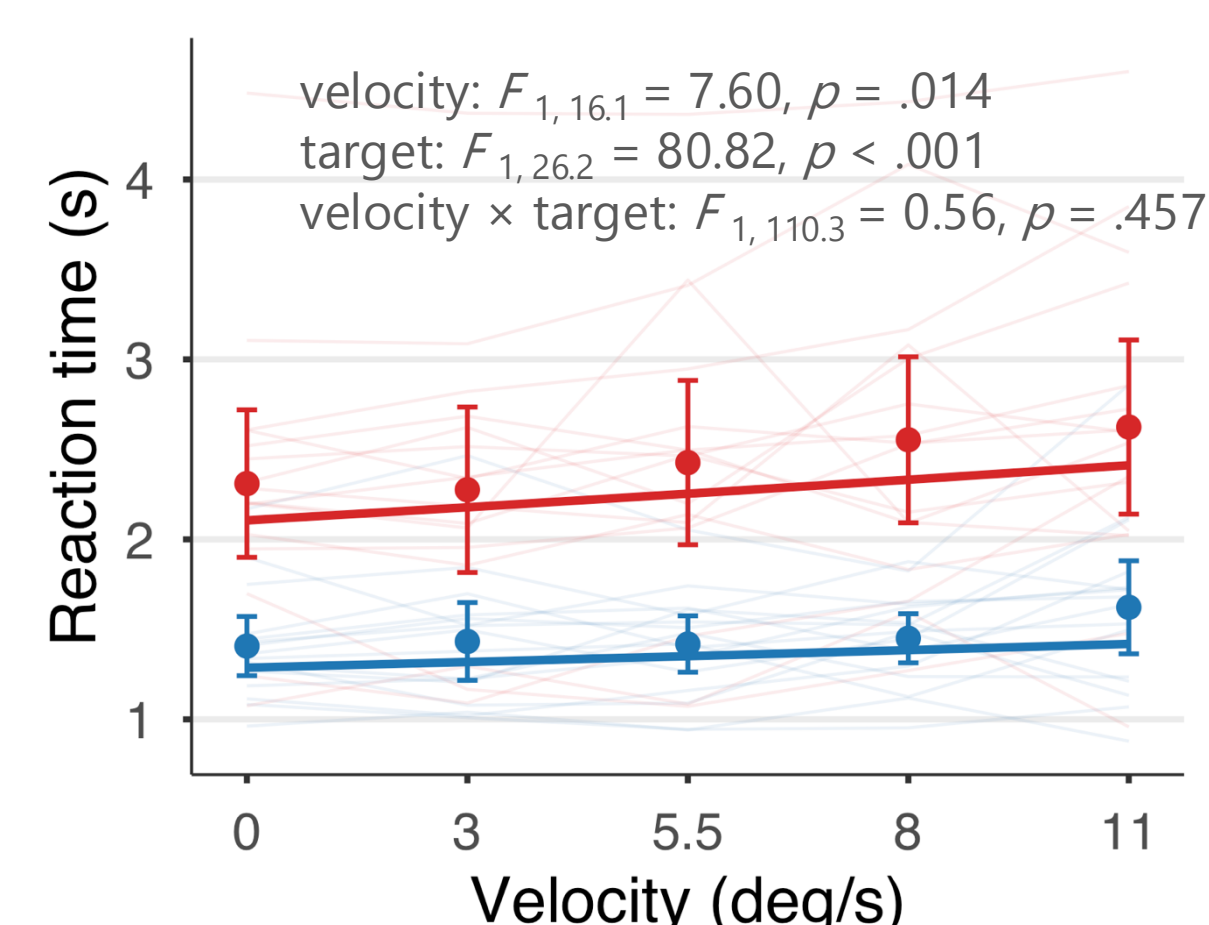
We investigated how **velocity** affects eye movements and behaviour in **visual search**, and whether people rely on adaptive search strategies to cope with motion.

Dynamic visual search task (a) and Peripheral visibility task (b)

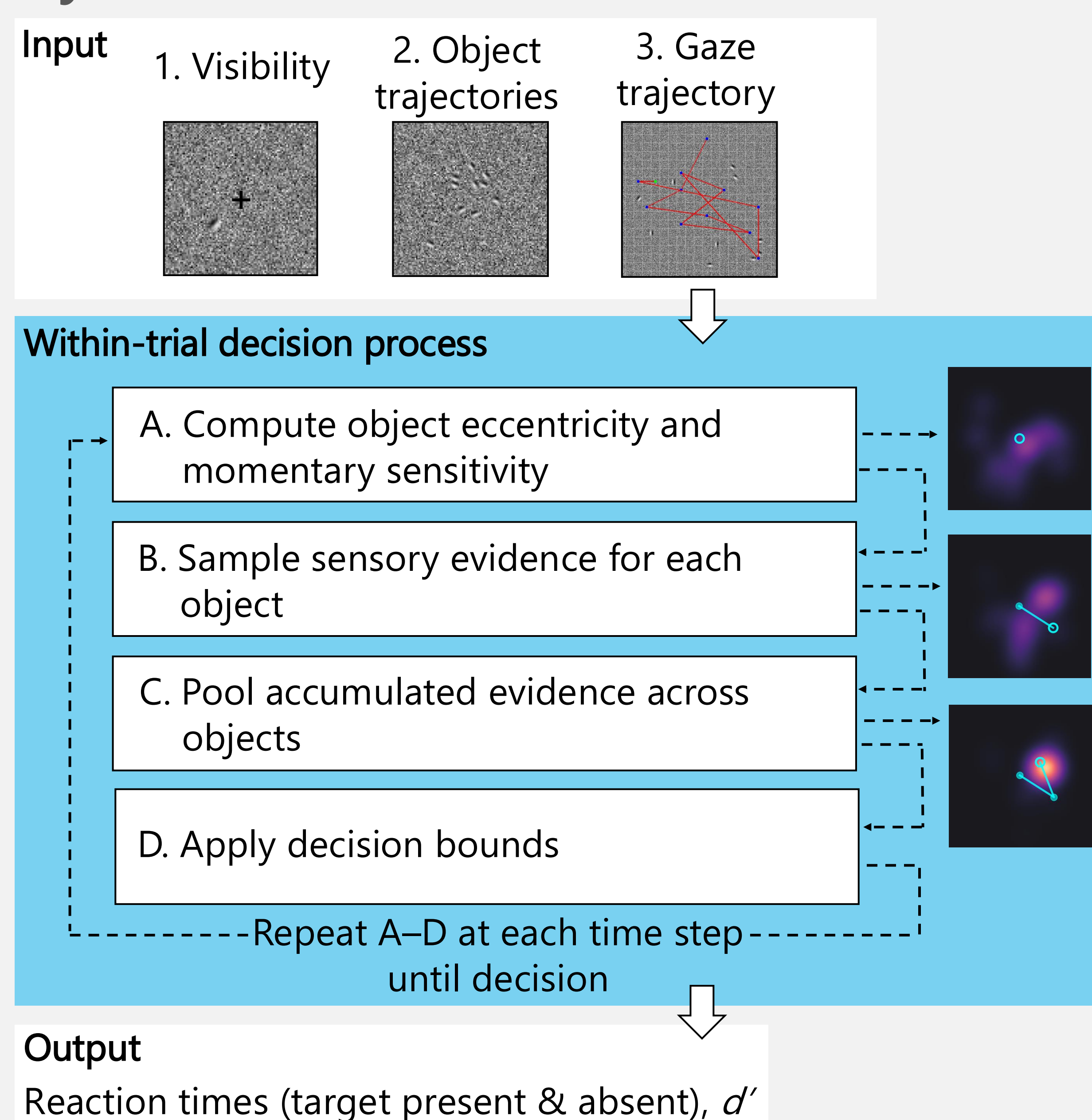


Results of the visual search experiment

Higher object velocities led to an increase in reaction times and a decline in accuracy, as well as more spatially constrained eye movements.

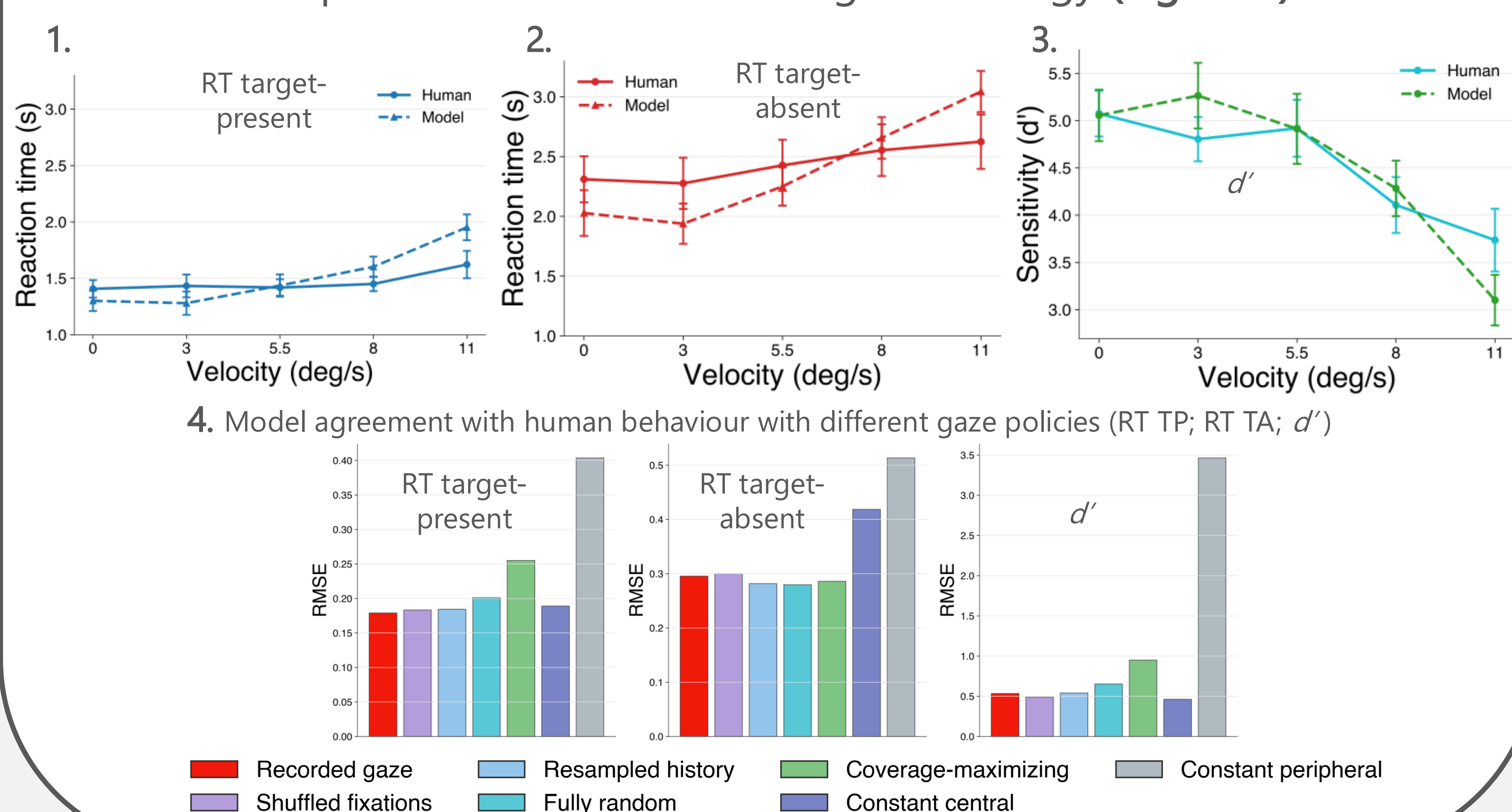


Dynamic Foveated Search Model



Model predictions

The DFSM closely predicted human performance in the search task (Figures 1, 2, 3). Results of temporally and spatially manipulating the gaze policy reveal that the model doesn't depend on the adaptiveness of the recorded gaze strategy (Figure 4).

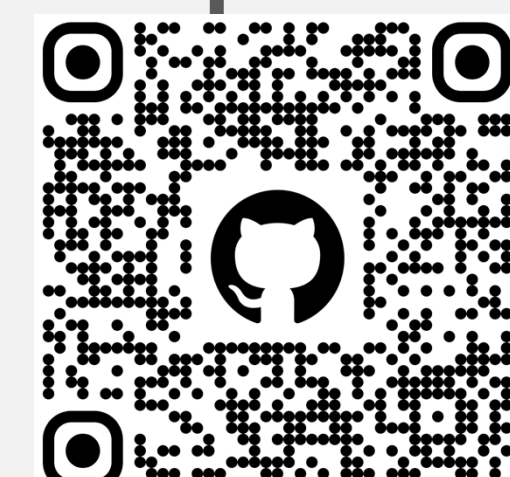


Conclusions

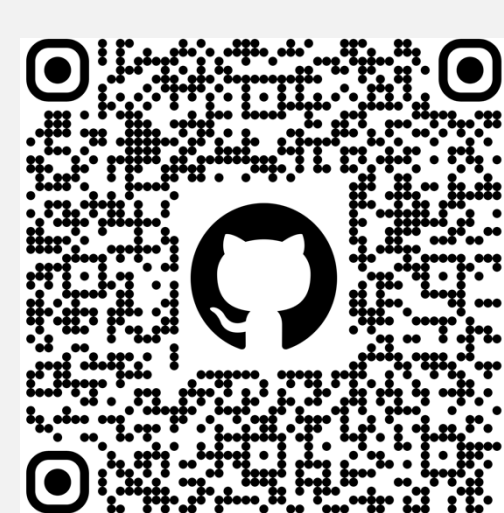
The decline in accuracy and increase in reaction times as object velocity increased can largely be explained by a straightforward Bayesian searcher with two free parameters (sensory noise & decision criterion), using replayed human gaze trajectories.

Perceptual constraints, rather than search strategy, play a dominant role in limiting performance in dynamic search.

Poster, data, scripts
and equations:



Stimulus demo



References:

Eckstein, M. P., Drescher, B. A., & Shimozaki, S. S. (2006). Attentional cues in real scenes, saccadic targeting, and Bayesian priors. *Psychological science*, 17(11), 973-980, 10.1111/j.1467-9280.2006.01815.x

Wolfe, J. M., Horowitz, T. S. (2017). Five factors that guide attention in visual search. *Nat. Hum. Behav.* 1, 1-8, <https://doi.org/10.1038/s41562-017-0058>